



8 (a) State the sum of the series $z + z^2 + \dots + z^n$, for $z \neq 1$. [1]

(b) By letting $z = \frac{1}{2}(\cos \theta + i \sin \theta)$ use de Moivre's theorem to deduce that

$$\sum_{m=1}^n \left(\frac{1}{2}\right)^m \sin m\theta = \frac{\left(\frac{1}{2}\right)^{n+2} \sin n\theta - \left(\frac{1}{2}\right)^{n+1} \sin(n+1)\theta + \frac{1}{2} \sin \theta}{\frac{5}{4} - \cos \theta}. \quad [6]$$

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(c) Use the result in (b) to find $\sum_{m=1}^n \left(\frac{1}{2}\right)^m m \cos m\theta$ in terms of n and θ . [You do not need to simplify your answer.] [3]

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(d) Hence find $\sum_{m=1}^{\infty} \left(\frac{1}{2}\right)^m m \cos m\theta$ in terms of $\cos \theta$. [You may assume that $\frac{n}{2^n} \rightarrow 0$ as $n \rightarrow \infty$.] [2]

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